

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

int cqueue[MAX];
int front = -1;
int rear = -1;

void enqueue(int val) {
    if ((rear + 1) % MAX == front) {
        printf("\nCircular Queue Overflow!\n");
    } else {
        if (front == -1) front = 0;
        rear = (rear + 1) % MAX;
        cqueue[rear] = val;
        printf("Successfully inserted %d\n", val);
    }
}

void dequeue() {
    if (front == -1) {
        printf("\nCircular Queue Underflow!\n");
    } else {
        printf("\nDeleted element: %d\n", cqueue[front]);
        if (front == rear) {
            front = -1;
            rear = -1;
        } else {
            front = (front + 1) % MAX;
        }
    }
}

void display() {
    if (front == -1) {
        printf("\nCircular Queue is empty!\n");
    } else {
        printf("\nCircular Queue elements: ");
        int i = front;
        while (1) {
            printf("%d ", cqueue[i]);
            if (i == rear) break;
            i = (i + 1) % MAX;
        }
        printf("\n");
    }
}

int main() {
    int choice, val;
    while (1) {
        printf("\n--- CIRCULAR QUEUE MENU ---\n");
        printf("1. Enqueue 2. Dequeue 3. Display 4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
```

```
--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 1
Enter value to insert: 100
Successfully inserted 100

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 1
Enter value to insert: 200
Successfully inserted 200

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 1
Enter value to insert: 300
Successfully inserted 300

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 3

Circular Queue elements: 100 200 300

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 2

Deleted element: 100

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 1
Enter value to insert: 400
Successfully inserted 400

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 3

Circular Queue elements: 200 300 400

--- CIRCULAR QUEUE MENU ---
1. Enqueue 2. Dequeue 3. Display 4. Exit
Enter choice: 4
Exiting program...
```